Amroha Police
Uttar Pradesh Police

AI DECISION SUPPORT FOR LAW ENFORCEMENT

FIR Assistant

Functional Overview and Operational Workflow

An AI-assisted cybercrime complaint classification tool that helps officers structure complaints, identify relevant legal provisions, plan evidence collection, and generate assessment reports.

Project Functionality Document

Prepared for law-enforcement use

01 / PROJECT OVERVIEW

Purpose of FIR Assistant

FIR Assistant supports the initial assessment of cybercrime complaints received by police personnel. It converts an unstructured victim statement into a consistent, reviewable complaint analysis.

> **Primary objective:** reduce the time required for preliminary complaint screening while helping officers identify offence categories, possible FIR provisions, missing facts, and relevant digital evidence.

Target Users

Police Station Personnel

For structured intake of walk-in, written, or digitally received cybercrime complaints.

Cyber Cell Officers

For preliminary legal mapping, evidence planning, and review before further action.

Investigating Officers

For quickly identifying missing facts and digital records that may require preservation.

Supervisory Officers

For reviewing a consistent summary and downloadable assessment report.

Supported Complaint Types

Financial fraud · Digital arrest scam · Phishing · Identity theft · Account takeover · Cyber stalking · Online threats · Data theft · Fake profiles · Intimate-image abuse

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FIR Assistant is a decision-support tool. It does not replace the legal judgment of the authorised officer, prosecutor, magistrate, or court.

02 / OPERATIONAL WORKFLOW

How the Application Works

1. Enter Complaint	2. Select Language	3. AI Analysis	4. Review Sections	5. Download Report

1

Complaint Intake

The officer types or pastes the victim statement. Input validation requires a meaningful description and limits excessively long submissions.

2

Language Selection

The officer selects English or Hindi. The complaint-entry interface and generated analysis/report follow the selected language.

3

Meaning-Based Classification

The complaint is sent securely to the configured Gemini model. The model analyzes context and conduct rather than relying only on keywords.

4

Structured Legal Output

The API returns validated JSON containing category, severity, confidence, legal suggestions, missing facts, evidence, and actions.

5

Officer Review

The result is displayed as an operational report. Suggested sections can be reviewed against complete facts and current law.

6

Report Download

A self-contained HTML report can be downloaded, opened in a browser, printed, or saved as a PDF for case documentation.

User Feedback During Processing

- Animated analysis state communicates that the AI request is in progress.
- Offline detection disables classification and displays a clear connection message.
- Invalid server responses are converted into understandable errors instead of technical parser messages.
- Model fallback handles temporary demand or rate-limit failures.

Analysis Output

Module	Function	Operational Benefit
Complaint Summary	Creates a concise factual summary of the victim statement.	Speeds up review and briefing.
Primary Category	Identifies the most suitable cybercrime category and secondary categories.	Supports consistent complaint routing.
Severity	Returns Low, Medium, High, or Critical based on supplied facts.	Helps prioritize initial attention.
Suggested FIR Sections	Maps facts to relevant IT Act, 2000 and BNS, 2023 provisions with reasons.	Provides a starting point for legal review.
Missing Facts	Lists information required before final section selection.	Improves statement completeness.
Evidence Checklist	Suggests screenshots, transaction details, call records, links, devices, and other records.	Supports timely evidence preservation.
Immediate Actions	Provides practical first-response steps appropriate to the complaint.	Reduces delay in urgent cases.
Downloadable Report	Generates an English or Hindi assessment document.	Creates a portable review artifact.

Legal Knowledge Context

The AI receives a curated legal reference covering commonly relevant provisions, including IT Act sections 43, 65, 66, 66B, 66C, 66D, 66E, 66F, 67, 67A, 67B, 70, 72, and 72A, together with commonly paired BNS provisions for cheating, personation, extortion, forgery, criminal intimidation, stalking, and related conduct.

04 / TECHNOLOGY AND SAFEGUARDS

System Architecture

React Frontend

Responsive police-branded interface, language toggle, loading feedback, result presentation, and report generation.

Node.js / Express API

Validates complaints, protects API credentials, constructs prompts, handles model fallback, and returns structured results.

Google Gemini

Performs contextual complaint classification and generates legal decision-support content.

Deployment

Supports a single Node service on Render/Railway and serverless frontend/API deployment on Vercel.

Reliability and Safety Controls

- API keys remain server-side and are excluded from Git through `.gitignore`.
- Strict structured-output schema reduces malformed AI responses.
- Fallback models are used for temporary overload or malformed output.
- Request timeout prevents the interface from waiting indefinitely.
- Complaint length validation reduces accidental oversized requests.
- Downloaded report content is HTML-escaped to prevent injected markup.

Important Operational Limitations

> AI confidence is a model self-estimate, not a calibrated probability. Suggested legal provisions must be checked against the complete complaint, available evidence, current statutory text, applicable judgments, and local procedure.

Expected Law-Enforcement Impact

1. Faster preliminary screening of cybercrime complaints.
2. More consistent identification of complaint categories.
3. Earlier recognition of missing facts and digital evidence.
4. Better structured material for review by authorised officers.
5. Improved accessibility through English and Hindi support.